

If I had to restart this final project or the entire class, one of the biggest things I would do differently is dedicate more time to deeply learning the concepts and applications of Real-Time Systems (RTS). During this summer, I was unable to dedicate as much time as I would have liked to learning RTS because I was also working as a CWEP at Lockheed Martin. I needed to get as many hours as possible since I must take reduced hours in the fall for Senior Design I, which will affect my ability to pay rent. Despite my lower involvement in the class, I still want to continue learning RTS as I find the subject matter incredibly interesting. I even bought an ESP32-S3 starter kit at the beginning of the summer thinking it would be a fun way to engage with the material. Unfortunately, I haven't even opened the box since the day it arrived. Additionally, I would choose to take this course in person. I personally struggle with online classes because I find it hard to maintain focus while staring at a screen for hours. If given the opportunity to take this class in person with Dr. Mike, I would jump at the chance.

I would say that the hardest part of this class was the practical applications. Even though I had taken Embedded Systems the semester right before this one, I still felt incredibly lost when it came to most of the assignments, especially all the code required for the web server. I will admit that it took help from a student who had previously taken the class, fellow students on Yellowdig, and assistance from LLMs just to get the HTTP server working for Application 1. This final project similarly required significant help from GenAI and LLMs for both the web server code and the GitHub components. Despite the difficulty, I greatly appreciate the patience and understanding Dr. Mike showed us, providing so much support through the scaffolding and adjusting the pacing to help us succeed.

The most valuable thing I learned throughout this class was surprisingly not a technical concept or a coding skill, but something far more personal: my mindset. At the beginning of the semester, I genuinely contemplated dropping this course because the workload, concepts, and projects felt so daunting. To be honest, I was afraid. Entering one of my last three semesters, I had never felt so lost in a class before. I have had trouble understanding isolated concepts in courses like Semiconductors or EMF, but never entire projects like Application 1. Beyond recognizing basic C code, I knew nothing about RTS, HTTP servers, or GitHub. Standing here right now writing this reflection with the real possibility of earning an A in this class feels unbelievable. If I could

tell my past self at the start of the summer what I would accomplish, both in this class and over the entire summer, I honestly wouldn't have believed it.

My mindset throughout what felt like a blur of a summer semester has completely transformed. I used to constantly think of exciting projects like building my own heart rate monitor watch, an arcade game emulator, or an RC car but I never had the drive to actually pursue them, making excuses like *"I don't have the skills," "I don't have the time,"* or *"I'll do it later."* Now, with only two semesters left in my degree and almost no side projects to my name, this class backed me into a corner and forced me out of my comfort zone. It made me realize that the only thing stopping me from doing these fun, exciting things was myself. With three weeks left before the start of the fall semester, I fully intend to review the course material, put real effort into mastering RTS, build cool projects with my ESP32, and finally create something of my own. Thank you, Dr. Mike, for giving me an incredible opportunity to grow not only in my technical skills, but in my mindset as well.